GROUP NUMBER: Student Group 9909

FULL LEGAL NAME	LOCATION (COUNTRY)	EMAIL ADDRESS	MARK X FOR ANY NON-CONTRIBUTING MEMBER
RUTHVIK MOHAN MENDUGUTTALA	INDIA	ruthvikmohan007@gmail.com	
SHIJIE DING	CHINA	dingshijie777@outlook.com	

Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
Team member 1	**RUTHVIK MOHAN MENDUGUTTALA**
Team member 2	**SHIJIE DING**
Team member 3	

Use the box below to explain any attempts to reach out to a non-contributing member. Type (N/A) if all members contributed. **Note:** You may be required to provide proof of your outreach to non-contributing members upon request.
N/A

Tasks

1: Data Quality

a.

A survey data where some entries have missing phone numbers, dates of birth stored in inconsistent formats (e.g., "01/02/2023" vs. "February 1, 2023"), and duplicate entries for the same customer with conflicting information (e.g., different email addresses). Another example can be a financial data where the Market Cap field contains NULL for 30% of records; Currency column mixes formats (e.g., "USD", "$", "US Dollar")

b.

The dataset above lacks completeness due to missing phone numbers, violates consistency with mismatched date formats, and fails accuracy because duplicate records contain conflicting data. Additionally, the lack of standardization makes it difficult to validate integrity, as duplicates cannot be reliably consolidated without risking data loss. For the financial data, missing market cap values violate data completeness requirements. Multiple currency formats break standardization rules, preventing reliable aggregation.

c.

A collection of social media posts labeled as "positive sentiment" that includes irrelevant content, like slangs, emojis, spam messages, non-text data like images without captions), mixed languages within a single post, slang, and text riddled with typos.

d.

This dataset fails relevance due to unrelated or meaningless content (spam), lacks consistency from mixed languages and formatting, and undermines accuracy with typos and slang. The absence of clear structure or metadata makes it challenging to filter noise, validate sentiment label.

2: Yield Curve Modeling

a. **Select Government Securities from India**

We use Indian Government Bonds (G-Secs) with the following maturities and yields (as of July 4, 2025) from Investing.com:

Maturity	Yield (%)
6M	5.43
1Y	5.47
2Y	5.68
5Y	6.04
10Y	6.29
20Y	6.81
30Y	7.01

b–d. Fit Nelson-Siegel and Cubic Spline Models

Nelson-Siegel Model

The Nelson-Siegel model is defined as:

$$y(\tau) = \beta_0 + \beta_1\left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau}\right) + \beta_2\left(\frac{1 - e^{-\lambda\tau}}{\lambda\tau} - e^{-\lambda\tau}\right)$$

Where:

- β0 (or) beta 0: long-term level
- β1 (or) beta 1: short-term slope
- β2 (or) beta 2: medium-term curvature
- λ (or) lambda: decay factor

We use nonlinear least squares to estimate parameters.

Cubic Spline Model

Cubic spline interpolation fits a smooth curve through all data points using piecewise third-degree polynomials. It ensures continuity and smoothness at the knots (maturities).

e. Compare the Models

Feature	**Nelson-Siegel**	**Cubic Spline**
Type	Parametric	Non-Parametric
Fit	May not pass through all points	Passes through all data points
Interpretation	Economically meaningful parameters	Purely Mathematical
Flexibility	Moderate	High (Depends on number of knots)
Risk of overfitting	Low	Higher if too many knots

f. Model Parameters (Example Output)

Assume we estimated the following for Nelson-Siegel:

- $\beta 0 = 6.85$
- $\beta 1 = -1.20$
- $\beta 2 = 0.75$
- $\lambda = 0.45$

These imply:

- Long-term rates stabilize around 6.85%
- Short-term rates are declining (negative slope)
- There's a moderate hump in the medium term

g. Is Nelson-Siegel Smoothing Unethical?

Smoothing becomes unethical when it misrepresents or hides the true nature of the data. In this case:

- **Nelson-Siegel is not unethical** because it is a well-established and transparent model used to summarize the term structure of interest rates.
- It does not fabricate data but provides a parsimonious and interpretable approximation.

- Ethical concerns arise only if smoothing is used to deceive or obscure risk, which is not the case here.

4: Empirical Analysis of ETFs

Daily returns are fundamental in financial analysis, they transform raw price data into a format that enables comparisons across different assets and time periods. Unlike prices, which often exhibit trends and non-stationarity, returns provide stationary. This transformation is essential for calculating risk metrics, as returns normalize scale differences between high-priced and low-priced stocks while capturing the relative day-to-day changes that investors experience. Daily returns is calculated as the percentage change in stock prices from one day to the next day. By examining daily returns, we can calculate volatility, and correlations among assets, which are essential steps for making informed investment decisions and constructing diversified portfolios.

Covariance matrix quantifies the degree to which the returns of different stocks move in relation to each other. A high covariance between two stocks suggests that they tend to rise and fall together. Using covariance matrix helps diversifying holdings to mitigate unsystematic risk and achieve their desired risk-return profile.

Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) are two techniques used to uncover the underlying structure within financial datasets. PCA transforms the original data into a new set of uncorrelated variables called principal components. These components are ordered by the amount of variance they explain in the data, with the first principal component accounting for the largest possible variance. PCA is useful for dimensionality reduction, enabling us to focus on the most significant factors driving market movements while discarding noise and less influential variables. The eigenvectors obtained from PCA represent the directions of these principal components in the original feature space, while the eigenvalues indicate the magnitude of variance explained by each component. A larger eigenvalue signifies a more important component in terms of capturing the data's variability. SVD, on the other hand, is a matrix factorization technique that decomposes a matrix into three separate matrices: U, Σ, and V^T. Here, U and V are orthogonal matrices containing the left and right singular vectors, respectively, and Σ is a diagonal matrix holding the singular values. The singular values in SVD provide a measure of the importance of each singular vector, similar to eigenvalues in PCA. However, the eigenvalues in PCA directly relate to the variance explained by each principal component, singular values in SVD represent the scaling factors for the singular vectors and can be used to determine the rank of the matrix or to approximate the matrix by retaining only the most significant singular values. Both techniques aim to extract meaningful information from the data by decomposing it into more interpretable components. However, PCA focuses specifically on finding the directions of maximum variance, whereas SVD provides a more comprehensive decomposition that can be utilized for multiple applications beyond variance explanation.

In the specific context, the eigenvectors from PCA highlight the combinations of stocks that contribute most to the overall variance in returns, potentially indicating common factors that drive the performance of these tech companies. For example, a significant principal component might reflect the broader technology sector trend, with eigenvector loadings showing the relative influence of each stock on this trend. The corresponding eigenvalues quantify how much of the total variance in the dataset is attributable to each principal component, helping us understand the proportion of market movements explained by these factors. The left singular vectors (from matrix U) can be associated with the original observations (days in our case), indicating patterns in the returns over time. The right singular vectors (from matrix V^T) relate to the original variables (stocks), showing how each stock contributes to the latent factors. The singular values, displayed in matrix Σ, rank these factors by their significance, allowing us to focus on the most informative ones. By examining the SVD results, one can identify which stocks are heavily weighted in the most significant factors and how these factors evolve over the observed period.

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